Examiner only

6. Fluoride ions are added to drinking water in some areas of the UK.

The table below shows the effect of different levels of fluoride in drinking water on the number of decayed, missing and filled teeth (DMFT) and the percentage of people suffering from fluorosis.

Concentration of fluoride (mg/dm^3)	Mean DMFT	Percentage of people affected by fluorosis (%)
0.3	7.0	4
0.6	4.5	6
0.9	3.0	15
1.2	2.5	35
1.5	2.3	40
1.8	2.0	45
2.1	2.1	60
2.4	2.1	68
2.7	2.1	75

(a) The recommended concentration of fluoride ions to be added to drinking water in the UK is $1.0\,mg/dm^3$.

Use the information in the table to explain why this is the recommended level in terms of DMFT and fluorosis. [4]

DMFT

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Fluorosis

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Examiner only

(b) Chlorine is added to all drinking water in the UK, whilst fluoride is only added in some areas.

Explain why some people are opposed to adding fluoride but no-one is opposed to adding chlorine to drinking water. [2]

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(c) Fluorine reacts with titanium at very high temperatures, forming titanium fluoride.

In an experiment 3.8 g of fluorine reacted with 2.4 g of titanium.

Calculate the simplest formula for titanium fluoride. You **must** show your working. [3]

$A_r(Ti) = 48$ $A_r(F) = 19$

Simplest formula ....

9

